

Jiachi ZHANG

Tel: +86 15816658985 **Email:** jingzhizhang569@gmail.com **GitHub:** github.com/tyui99

Web: tyui99.github.io **Google Scholar:**
scholar.google.com/citations?user=4BDSTP8AAAAJ&hl=en

EDUCATION

South China University of Technology 09/2023 - 06/2027

Qualification: *Bachelor of Engineering*

Major: *Information Engineering*

Average Score: 84.52/100

Core Courses: *Signals & Systems, Analog Electronics, Digital Electronics, Digital Signal Processing, Principle of Communications, Microcomputer System and Interface Technology, Programming in C++*

SUMMER STUDY PROGRAM

École Polytechnique de l'Université de Nantes, France 07/2025

Core Courses: *Français, Théorie de l'Information*

INTERNSHIP EXPERIENCE

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences 10/2025 - Present

Position: Visiting Research Intern (Advisors: Dr. Wenqi Fang & PhD candidate Zhuoyu Wu)

Carried out independent research by identifying research gaps, developing research hypotheses, designing experimental protocols, and conducting hands-on experiments and data analysis.

Compiled academic materials by reviewing relevant literature and drafting concise and structured experimental summaries.

Authored and revised the thesis, ensuring a clear, logically rigorous scientific narrative grounded in experimental results.

PUBLICATIONS & RESEARCH

Gain-Aware Prediction-Space Recursive Controller for Lightweight Polyp Segmentation 09/2025 - Present

Submitted to ICONIP 2026.

Formulated segmentation refinement as a prediction-space recursive correction task, shifting correction entirely into logit space to avoid re-engaging heavy feature hierarchies.

Designed a three-stage recursive controller (Prediction Evidence Encoding → Gain-Aware State Update → State-Guided Residual Correction) with only 189 parameters and negligible GMACs overhead.

Evaluated across 7 lightweight backbones (UltraLBM-UNet, Mobile-PolypNet, EGE-UNet, CGNet, ULite, CMUNeXt-S, I2U-Net) on 4 polyp segmentation datasets under a unified Kvasir-trained protocol.

Achieved consistent source-domain improvements and competitive performance against both training-side baselines (+LL/+DS/+Mutation/+LoMix) and the heavier structural refiner +Harmonizing.

RIGS-Refiner: Risk-Guided Recursive Refinement in Prediction Space for Polyp Segmentation 10/2025 - Present

Submitted to APSIPA ASC 2026.

Proposed a lightweight post-refinement plugin for risk-guided recursive correction directly in prediction space, built on frozen host predictions.

Designed a shared recursive cell combining image priors (Sobel edge, average pooling), prediction cues, risk-guided update, and residual correction with only 519 parameters and +0.631 GFLOPs.

Tested on PraNet and SegFormer-B0 across Kvasir-SEG, ClinicDB, ColonDB, and ETIS, achieving consistent IoU/HD95/cIDice improvements with a favorable efficiency–accuracy trade-off against SegFix, rNCA, and CascadePSP.

Glomerular Pathological Image Detection via YOLO11 with Stain Normalization 04/2024 - 03/2025

Designed color clustering and color transfer modules to mitigate staining variance and augment glomerular pathological image datasets.

Constructed a YOLO11-based detection framework and adopted Focal Loss to balance positive-and-negative samples in the glomerular detection task.

Integrated EMA multi-scale attention to enhance feature fusion and suppress tissue noise; validated via ablation experiments, achieving 94% mAP that outperformed baseline models.

PROJECT EXPERIENCE

Macroeconomic Indicator Forecasting System via SAITS and Time-Series LLMs 07/2025 - 08/2025

Adopted the SAITS algorithm to conduct multivariate time-series imputation, and reconstructed high-fidelity continuous feature spaces under limited labeled data.

Integrated TimeLLM and ChatTS, designed cross-modal alignment modules to capture temporal evolution trends and enhance prediction accuracy.

Constructed a modular training and evaluation system, and developed high-precision interpretable time-series models, securing top 10 in the E Fund FinTech Challenge.

FPGA-Based Shooting Confrontation Game 03/2026

Developed a dual-mode FPGA shooting game supporting 1280×800@60fps HDMI output, infrared control and real-time score display.

Designed HDMI display drivers and SDRAM caching logic, and implemented AABB-based collision detection for game objects.

Used finite state machines to control seven game states and ensure synchronization between peripherals and display modules.

Completed project validation and received the Excellent Course Design award for outstanding academic performance.

EXTRACURRICULAR EXPERIENCE

President, Art Troupe, Youth League Committee 11/2024 - 11/2025

Participated in the production and singing of the department song, and performed in the college band on various occasions.

Managed and coordinated internal departmental affairs, and took charge of the college graduation gala as a key organizer for multiple times.

Study Representative, Class Committee 09/2023 - 10/2025

Collected and distributed daily assignments, and completed the liaison work for academic affairs.

Communicated actively between teachers and classmates, and released official university notifications in a timely manner.

HONORS & AWARDS

3rd Prize Scholarship, South China University of Technology 12/2025

Merit Student, South China University of Technology 12/2025

Advanced Individual in Student Work, South China University of Technology 09/2025

Active Participant in Student Work, South China University of Technology 09/2025

10th Place, Pazhou Algorithm Competition - E Fund Cup FinTech Challenge 08/2025

Outstanding League Cadre, South China University of Technology 05/2025

ADDITIONAL SKILLS

Language Ability: French (basic), English (fluent, can be used as a studying language)

Programming & Development: Python, PyTorch, Keil, MATLAB, Vivado